

Arunabha Majumder

Mechanical & Robotics R&D Engineer — Actuator Development, Controls & Hardware Systems

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PROFESSIONAL SUMMARY

Mechatronics and Mechanical R&D Engineer with practical experience in product development, designing, prototyping and testing dynamic electromechanical systems, variable-stiffness actuators and robotic mechanisms. Proven ability to bridge CAD modelling, kinematic analysis, control implementation and sensor integration into functional hardware solutions.

WORK EXPERIENCE

Research Engineer — Robotics & Actuator Development (PhD Project: VIE XO) | 2023 – Present *Aalborg University, Aalborg, Denmark*

- Cut shoulder exoskeleton energy consumption by more than 25% by designing a hybrid actuation architecture that coordinates a parallel spring with a 6 Nm motor, offloading quasi-static gravity torque across the assistance cycle; won the Best Paper Award, IFToMM ISRM 2026.
- Reduced user anterior deltoid muscle effort by more than 15% by deploying an Assist-as-Needed (AAN) hybrid controller on functional prototype hardware, tested with human participants.
- Developed a sensorless end-effector force-estimation method for a 5-bar planar parallel robot using variable-stiffness actuators (VSA) and the robot Jacobian, validated against a force sensor to below 10% RMS error, and used it to measure human arm impedance in the gait lab.
- Conducted data collection for a 12-participant study across 3 assistance modes and 2 load conditions, using EMG and motion-capture instrumentation.
- Built the full prototype stack from concept to tested hardware: SolidWorks CAD and FEA, mechanical fabrication, motor, encoder, force-sensor and DAQ integration, embedded control and custom automated test rigs.
- Taught university students and supervised Bachelor's (B.Tech) students (2023 – 2025).

M.Tech Researcher — Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI — National Laboratory, Govt. of India), Durgapur, India

- Reduced dynamic position-tracking error to 0.3–0.78% by designing and implementing a neural-network-based gain-scheduled PID controller for a highly non-linear pneumatic actuator, outperforming the classical PID baseline under varying axial load.
- Fabricated pneumatic artificial muscle actuators in-house from raw materials and built the test set-up with valves, pressure sensors and DAQ to characterise load–displacement behaviour and derive an empirical actuator model.

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

SKILLS & INTERESTS

- **Design & Modelling:** Mechanical Product Development, Actuator & Robotic Mechanisms, CAD, FEA, Design for Assembly (DFA), 3D Printing
- **Robotics & Controls:** Compliant Actuation, Kinematics, Jacobian-Based Force Estimation, Control Systems, Automation, Pneumatic Control

- **Testing & Hardware:** Prototype Fabrication, Sensor Integration, Custom Test-Rig Development, Data Collection, Qualisys Motion Capture, EMG, Forsentec Force/Torque Sensors
- **Software & Programming:** SolidWorks, Fusion 360, Ultimaker Cura, MATLAB/Simulink, Python, LabVIEW (basic), Arduino, LaTeX, Adobe Illustrator, Adobe Photoshop, Procreate (basic)
- **Languages:** English (CEFR: C1), Bengali (Native), Hindi (oral proficiency)
- **Interests:** Engineering Design & Prototyping, Visual Arts & Illustration

PROJECTS

Rehabilitation Robotics (Bachelor's Capstone) | 2018 – 2019

Siddaganga Institute of Technology, Tumakuru, India

- Delivered an award-winning upper-limb rehabilitation exoskeleton that secured the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant, leading CAD design, fabrication and user testing end to end. Video: youtube.com/watch?v=q5Ystz9NCpQ

Engineering Design & Prototyping (Extracurricular) | 2017 – 2018

Siddaganga Institute of Technology

- Led end-to-end development of a custom-fabricated Go-Kart from the ground up (2017).
- Collaborated in a multidisciplinary team to engineer an eco-friendly human-electric hybrid vehicle for the Efficycle competition (2018).

CERTIFICATIONS

- Certified SolidWorks Associate (CSWA) — Mechanical Design

ACHIEVEMENTS

- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics — awarded for hybrid-actuated exoskeleton design (2026)
- Best Major Project Award, Department of Mechanical Engineering (2019)
- TATA Technologies Innovation Award — INR 100,000 Grant (2019)

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026, MMS*, vol. 213, Springer, 2026.
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

VISUAL ARTS & ILLUSTRATION

- Professionally trained visual artist holding a Diploma in Fine Arts (Painting) from Bangiya Sangeet Parishad.
- Use sketching, painting and graphic design skills to create illustrations and technical visuals.

REFERENCES

Available upon request.